Closed-loop DC heat dispatch for a 500 m³ pool (v04)

Executed Stateflow and Simulink R2026a: 168 hours, eight synthetic scenarios. Not field-measured.

1. Architecture and physical scope

A new closed-loop dispatcher receives water temperature, pool heat demand and TES inventory as feedback. Stateflow selects direct exchanger, heat pump, TES-only, IT shutdown or no-source operation. The unchanged original v85 is a preserved reference, not the controller of this new v04. DC heat input is generated synthetically.

2. Thermodynamic conservation and storage

The pool is modelled as one well-mixed thermal capacity, with assumed evaporation and heat-loss coefficients. TES is a lumped high-grade store without stratification or hydraulic sizing. COP(T) uses a bounded Carnot-style hypothesis, not verified manufacturer performance.

$$C_{pool} = \rho V c_p = 997 \cdot 500 \cdot 4182 = 2\,084\,727\,000 \text{ J/K}$$

$$E_{water}(16 \rightarrow 28°C) = C_{pool} \cdot 12 / 3.6e9 = 6.94909 \text{ MWh (without losses)}$$

$$T(k+1) = T(k) + 900 \, (Q_{pool} - Q_{loss}) / C_{pool}$$

$$Q_{loss} = Q_{evap} + Q_{conv} + Q_{rad} + Q_{construction} + Q_{makeup} + 2 \text{ kW}$$

3. Allocation and protective constraints

Thermostat: ON below 27.35 °C, OFF above 27.65 °C, predicted water ceiling 27.75 °C. IT_ok=0 routes recoverable heat to independent rejection, while TES may still serve district heat. BAZEN_ok=0 stops pool heating only. Heat-pump output is capped at 200 kW, compressor at 55 kW, pool input at 150 kW.

$$Q_{DC,available} = Q_{direct} + Q_{HP,source} + Q_{reject}$$

$$Q_{HP,out} = Q_{HP,pool} + Q_{HP,DH} + Q_{HP,TES} = Q_{HP,source} + P_{HP,compressor}$$

$$Q_{pool} = Q_{direct} + Q_{HP,pool} \; ; \; Q_{DH} = Q_{HP,DH} + Q_{TES,discharge}$$

$$E_{TES}(k+1) = E_{TES}(k) + 900\,(0.93\,Q_{charge} - Q_{discharge}/0.91 - Q_{standing,loss})$$

$$Q_{pool} \le \min(150\ \text{kW}, Q_{pool,requested}, Q_{safe}) \; ; \; Q_{safe} = \max(0, C(27.75-T)/900 + Q_{loss})$$

$$COP_{scenario} = \text{clip}(0.43 \cdot (52+273.15)/(52-(T_{DC}-5)), 2.5, 6)$$

4. Executed comparison scenarios

Scenario	Pool MWh	DH MWh	HP el. MWh	Unmet DH MWh
Base	10.39	16.10	4.76	1.54
Pool disabled	10.02	16.10	4.82	1.54
IT shutdown 24h	10.31	14.57	4.56	3.07
No DC source	10.31	14.57	4.56	3.07
Warmer source	10.40	15.08	2.74	2.56
No pool cover	11.41	16.17	5.00	1.47
TES disabled	10.39	15.61	4.58	2.03
High DH demand	10.39	22.88	6.00	10.72

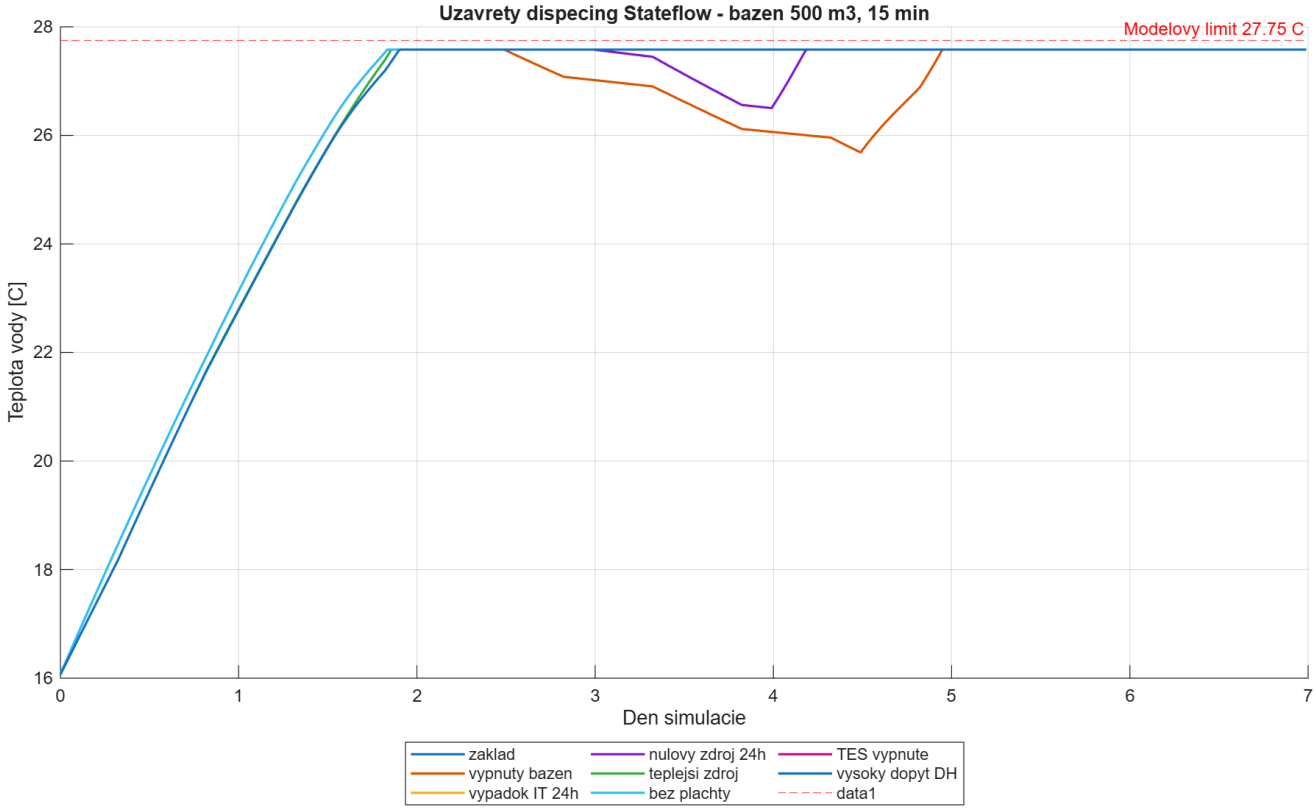
5. Independent QA, uncertainty and next work

Independent CSV analysis passed 65493 checks for eight cases. For every 15-minute time step it verifies source and heat-pump first-law balances, TES energy, pool temperature, Stateflow mode and emergency isolation. No physical commissioning or certified compliance is claimed. Measurements, manufacturer COP maps, hydraulic sizing, natatorium HVAC, fault-tolerant sensing and consolidated regulatory review remain necessary.

Maximum modelled T = 27.58°C; 8 × 672 steps of 900 s.

Original v85 SHA256: 2a9de22c0c55e8c4a87f7b0e0cfe9cd97db08c991f09d98361b009de310adde7

6. Pool temperature during the seven-day scenarios



Method: MATLAB model, CSV exports, independent Python QA; operating inputs class D = assumed synthetic.
EU 2024/1364: https://eur-lex.europa.eu/eli/reg_del/2024/1364/oj
EN 15288-1+A1:2024, swimming pool safety; applicability and actual law must be independently checked.